CHETAN BORSE

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EDUCATION

M.S., Robotics and Autonomous Systems Graduating May 2026
Arizona State University, Tempe, AZ 4.0 GPA
Ira Fulton School of Engineering
Relevant coursework: Robot Modeling & Control, Perception, Reinforcement Learning, Multi-Robot Systems

TECHNICAL SKILLS

Simulation & Frameworks: MuJoCo, MuJoCo-Warp, GazeboSim, ROS, NVIDIA Warp, PyTorch
Programming: Python, MATLAB, C++, Bash, Arduino
Robot Hardware: LEAP Hand, Wonik Allegro Hand, Franka Emika Panda
Design/Manufacturing: SolidWorks, Fusion360, Cura, 3D Printing

PUBLICATIONS

ComFree-Sim: A GPU-Parallelized Analytical Contact Physics Engine for Contact-Rich Robotics Simulation
Chetan Borse, Zhixian Xie, Wei-Cheng Huang, Wanxin Jin
Submitted to Intelligent Robots and Systems (IROS) 2026, under review [Website] [Paper]

RESEARCH

IRIS Lab, Tempe, AZ: Master's Thesis Student August 2025 – Present

- Developed ComFree-Sim, a GPU-parallelized analytical contact physics engine for scalable contact-rich robotics simulation with upto 2-3x throughput compared to MuJoCo-Warp.
- Leveraged ComFree-Sim for sampling-based MPC (MPPI) for dexterous manipulation tasks.
- Deployed control policies on LEAP Hand hardware, achieving stable real-time in-hand manipulation at $\sim$20 Hz.

PROJECTS

Generalizing Object Manipulation using Goal-Conditioned RL [Code] October 2025 - December 2025

- Reproduced key results from ECRL paper using Transformer-based actor-critic policies (TD3+HER) with entity-centric representations.
- Trained models using Deep Latent Particles to turn raw pixels into structured object-level entities.
- Achieved strong compositional generalization for upto six objects after training for three objects.

Enhancing Visuomotor Diffusion Policy with AdaLN-Zero [Code] March 2025 - May 2025

- Improved training stability of diffusion policy networks by integrating AdaLN-Zero into Transformer architectures.
- Addressed instability and high computational cost in long-horizon visuomotor tasks.

Debris Detection using Swarm Robots [Code] October 2024 - December 2024

- Designed decentralized MATLAB-based mapping algorithm for multi-agent debris localization for surface vessels.
- Implemented distributed map-sharing to merge local maps into a unified global debris field map.

Path Planning for Autonomous Underwater Vehicle (AUV) [Code] March 2025 - May 2025

- Simulated AUV navigation in GPS-denied environments using ROS Gazebo (UUV Simulator).
- Implemented A* planning with LiDAR-based obstacle detection and occupancy grid mapping.
- Fused DVL and IMU data for odometry estimation.

EXPERIENCE

Lycan Aerospace, Bangalore, India: Design Engineer April 2023 – May 2024

- Designed and prototyped structural and payload systems for a 10L-capacity agricultural UAV (AGROS), producing manufacturing drawings and collaborating cross-functionally to ensure manufacturability and system integration.

Tata Consultancy Services, Mumbai, India: Systems Engineer October 2021 – May 2022

- Supported migration from legacy mainframe systems to an AWS-based distributed Java environment, monitoring batch processes and resolving data discrepancies in production pipelines.